

Evolved biological factories: constraint-based evolutionary search for robust metabolic architectures in *E. coli*

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1 Introduction

Engineered microbes convert renewable feedstocks into fuels, materials, and chemicals, but the design of production strains remains guided primarily by human intuition: an engineer picks a pathway, expresses it, and hopes the host tolerates it. An alternative is to let a search process explore the space of possible metabolic architectures and select for what works, under conditions that resemble an industrial process rather than a single idealized optimum. This project asks whether a computational evolutionary search, starting from a target product rather than an existing pathway, can discover non-obvious biological architectures that match or exceed known engineered pathways under multi-condition robustness criteria. The design principles that let engineered cells maintain high productivity while remaining metabolically stable are the scientific object of interest; the evolved architecture library is the deliverable.

Three hypotheses are pre-registered (PRE-REGISTRATION.md, locked before any data or search): H1 (performance) - at least one evolved architecture reaches predicted product flux at or above the published engineered pathway evaluated in the same genome-scale model under identical uptake bounds; H2 (novelty) - the evolved library contains architectures absent from the published pathway that are mechanistically interpretable; H3 (robustness principle) - architectures selected under multi-condition pressure retain their production across a locked stress battery at a significantly higher rate than architectures selected under single-condition pressure. Honest negatives are preserved and redirect the project internally; no negative is published as the final result.

2 Pre-registered design

2.1 Hosts, models, and targets

The primary host is *Escherichia coli* K-12 MG1655 in the genome-scale metabolic model iML1515, with the yeast arm (Yeast8) reserved for cross-host robustness analysis. Three locked target products span distinct chemistry classes: 1,4-butanediol (non-natural industrial diol; benchmark route Yim et al. 2011, the Genomatica pathway), isobutanol (branched-chain higher alcohol; benchmark Atsumi et al. 2008, the keto-acid route), and lycopene (tetraterpene; benchmark Alper et al. 2005, crtEBI with dxs push). Reaction lists per the cited papers were locked in data/benchmarks.json before any search, with DOI provenance.

2.2 Benchmark metric: production envelopes

For each target, the published engineered pathway was implemented in iML1515 under the locked base condition (M9 minimal, glucose 10 mmol/gDW/h, aerobic). Because production competes with biomass, a single max-growth number is meaningless: at 100% of maximum growth all three pathways carry zero product flux. The benchmark is therefore a production envelope: maximum product flux with growth fixed at locked fractions of the FBA maximum,

$$F_t(g) = \max_v v_{\text{product}} \quad \text{s.t.} \quad Sv = 0, v_{\text{biomass}} = g \cdot \mu_{\text{max}}, l \leq v \leq u, \quad (1)$$

evaluated at $g \in \{1.0, 0.9, 0.75, 0.5, 0.25, 0.0\}$ with a parsimonious-FBA tie-break. Measured envelopes (results/benchmark_flux.json): 1,4-BDO reaches $F = 10.35$ mmol/gDW/h at $g = 0$ and 5.27 at $g = 0.5$; isobutanol 10.00 and 5.24; lycopene 0.97 and 0.49. The envelope, not a single titer, is what an evolved architecture must beat at matched growth. Table 1 reports the measured envelopes for all three published routes at the six locked growth fractions; the FBA maximum growth rate of the host under the base condition is $\mu_{\text{max}} = 0.877 \text{ h}^{-1}$ for all targets (identical uptake bounds, wild-type core). The zero at $g = 1.0$ in every row is a verified property

Table 1: Measured production envelopes (product flux, mmol/gDW/h) of the three published engineered routes in iML1515 under the locked base condition, at growth fixed to fractions of μ_{max} (results/benchmark_flux.json, run 2026-09-26; pFBA tie-break coefficient 0.999).

Target route	$g=1.0$	0.9	0.75	0.5	0.25	0.0
1,4-BDO (Yim 2011)	0.000	1.060	2.649	5.267	7.878	10.355
Isobutanol (Atsumi 2008)	0.000	1.048	2.620	5.239	7.813	10.000
Lycopene (Alper 2005)	0.000	0.098	0.244	0.488	0.731	0.970

of the host–route system, not a solver artifact: biomass and product draw on the same carbon and redox budget, so the envelopes are strictly monotone decreasing in g , and any comparison at a single growth point would be meaningless without the matched-growth rule.

2.3 Search protocol

The tie-break inside each envelope point is parsimonious FBA: among optimal solutions, minimize total flux,

$$v^* = \arg \min_v \|v\|_1 \quad \text{s.t.} \quad Sv = 0, v_{\text{product}} = F_t(g), v_{\text{biomass}} = g \mu_{\text{max}}, l \leq v \leq u, \quad (2)$$

with the pFBA optimality slack fixed at 0.999; FBA and pFBA product fluxes were verified equal on all envelope points (unique optima), so the cheaper FBA is used for search fitness and full pFBA reserved for final evaluation. The search genome encodes on/off choices over a locked reaction pool (benchmark route blocks plus literature-noted alternatives, data/reaction_pool.json). Fitness is product flux at the pFBA-optimal solution with growth fixed at $g = 0.5$, the envelope midpoint. Search fitness is the nominal-condition envelope point at the locked growth fraction,

$$\Phi(a) = F_a^{(\text{nominal})}(0.5), \quad (3)$$

and architecture performance against the benchmark route is reported as the exceedance ratio

$$E_a = \frac{F_a^{(\text{nominal})}(0.5)}{F_{\text{benchmark}}^{(\text{nominal})}(0.5)}, \quad (4)$$

with $E_a > 1$ required for an H1 beat at matched uptake bounds. The exhaustive pools used so far have genome spaces of

$$|\mathcal{G}| = 2^h \times 2^t, \quad (5)$$

genomes per target (h host-level blocks, t target-specific blocks: $h=0, t=2$ in pool v1 giving 4; $h=4, t=2$ in pool v2 giving 64), so enumeration evaluates every genome exactly once and the stochastic search budget is dominated by construction. Significance of any evolved advantage is calibrated against a 1000-seed random-architecture null as the percentile

$$\hat{q} = \frac{1 + \sum_{s=1}^{1000} \mathbf{1}[\Phi(a_s^{\text{rand}}) \geq \Phi(a)]}{1001}. \quad (6)$$

The search is a $(\mu + \lambda)$ evolutionary algorithm ($\mu = 24$, $\lambda = 48$, tournament selection, 60 generations, five locked seeds), with a 1000-seed random-architecture null for significance calibration. On the version-1 pool (two alternative blocks per target, four genomes), exhaustive enumeration strictly dominates the stochastic search and was used instead; the EA scaffold is retained for expanded pools.

2.4 Stability battery

A scalar summary of production-growth coupling used in the envelope analysis is the local slope

$$\sigma_t(g) = \frac{F_t(g_2) - F_t(g_1)}{g_2 - g_1}, \quad (7)$$

the shadow price of growth in units of product flux per unit growth fraction; the measured envelopes are strictly monotone with $\sigma_t < 0$ throughout. Robustness is evaluated over a locked nine-condition battery (nutrient shifts, carbon shifts to glycerol and acetate, anaerobiosis, uptake perturbations, oxygen limitation). After a metric-interpretation bug was caught at first evaluation (growth referenced to the base-condition maximum, which made every condition untestable), the locked rule is growth fixed at 50% of the condition-specific maximum, with retention defined against nominal-condition flux. The nine locked conditions are: GLC_LOW (glucose uptake -2), GLC_MID (-5), GLC+20% (-12), GLC -20% (-8), GLYCEROL (sole carbon, uptake -10), ACETATE (sole carbon, uptake -10), ANAEROBIC (oxygen uptake 0), O₂ -50% (oxygen uptake halved), and the nominal base GLC_AEROBIC (glucose -10 , aerobic); all uptakes in mmol/gDW/h. Formally, for architecture a and condition c ,

$$R(a, c) = \frac{F_a^{(c)}(0.5)}{F_a^{(\text{nominal})}(0.5)}, \quad \text{BatteryPass}(a) = \bigwedge_{c \in \text{battery}} \mathbf{1}[R(a, c) \geq 0.8], \quad (8)$$

where $F_a^{(c)}(0.5)$ is the architecture's product flux in condition c with growth fixed at 50% of the condition-specific maximum. The pass bar of 0.8 was locked before any architecture was evaluated against the battery. What this battery measures is a model property - retention of modeled product-flux capability under perturbation - not, by itself, evidence of growth-coupled production robustness; the growth floor's status is examined per pool in the caveats.

3 Results

3.1 Single-condition search reproduces the benchmark envelopes

Across all fifteen seed runs (three targets, five seeds), the best evolved architecture exactly reaches the locked envelope value at $g = 0.5$: 5.27 mmol/gDW/h for 1,4-BDO, 5.24 for isobutanol, 0.49 for lycopene. The search recovers the published routes as optimal under single-condition selection - a validation of the evaluation pipeline, and the baseline for H1.

3.2 Search definitions (locked terminology)

To remove any ambiguity about what the method is: the *representation* is a genome vector of discrete, biologically typed block choices (h host-level levers and t target-specific route blocks over the locked reaction library); the *search policy* is either exhaustive enumeration of the genome space (strictly dominant at pool sizes ≤ 256 and used for all reported results) or a $(\mu + \lambda)$ evolutionary algorithm ($\mu = 24$, $\lambda = 48$, tournament selection, 60 generations, five locked seeds) retained for larger pools; the *objective* combines nominal-condition product flux against the matched-growth published-route envelope and the tiered robustness score R_{tiered} over the locked nine-condition battery; the *learning component* is none - no trained, fitted, or adaptive model appears anywhere in the pipeline, and no parameter is updated from data. The method is constraint-based evolutionary search over a genome-scale stoichiometric model, nothing more. Where a future amendment adds a learned surrogate or policy, it will be locked and named explicitly.

3.3 H2 candidate: a tail-only minimal 1,4-BDO architecture

A non-obvious architecture emerged from the search: with *both* heterologous upstream route blocks switched off, the four-step heterologous tail (4-hydroxybutyrate dehydrogenase, CoA transferase, CoA-acylating aldehyde dehydrogenase, alcohol dehydrogenase) still reaches the full nominal flux of 5.27 mmol/gDW/h. The model explains why: *E. coli*'s native GABA shunt (succinate-semialdehyde supply through glutamate metabolism) feeds the tail without any heterologous upstream enzyme. This tail-only architecture is absent from the Yim et al. pathway definition, uses four fewer heterologous enzymes, and is mechanistically interpretable - the H2 novelty pattern in its locked form at the nominal condition. As the tiered-battery subsection below shows, the candidate does not survive the battery, so the claim stands at nominal condition only.

3.4 H3 honest negative on pool v1, and a battery-design finding

Exhaustive multi-condition evaluation of pool v1 found that *no* genome passes the locked battery (80% retention in all nine conditions): best minimum retentions are 0.175 (1,4-BDO), 0.176 (isobutanol), and 0.27 (lycopene). Inspection shows the failure is structural, not a lack of search power: under the severe nutrient shift GLC_LOW (glucose at 2 rather than 10 mmol/gDW/h), elementary carbon balance caps a C4 product at roughly one to two mmol/gDW/h against a nominal 5.27, a retention ceiling of 0.2–0.3 that no architecture can lift. The bound is elementary: a C4 product made from glucose at uptake u_{glc} satisfies

$$v_{\text{product}} \leq \frac{6}{4} u_{\text{glc}} = 3.0 \text{ mmol/gDW/h at } u_{\text{glc}} = 2, \quad (9)$$

before any carbon is allocated to biomass, maintenance, or byproducts; with growth fixed at 50% of the condition maximum the achievable product flux falls to roughly one to two, giving $R(a, \text{GLC_LOW}) \leq 1.5/5.267 \approx 0.28$ for every conceivable architecture a - including the published routes and any architecture the search could ever return. The benchmark engineered routes fail identically. A pass bar of 80% retention on severe nutrient shifts is therefore unsatisfiable by construction, the battery provides no selection gradient, and H3 is untestable as locked. This is recorded as an honest negative with the design flaw named, and a redesign (retention against condition-achievable optimum rather than against nominal flux; tiered bars for shifts versus perturbations; robustness-oriented pool levers such as knockout blocks and co-utilization feeds) is being locked as a dated amendment before any re-evaluation, following the project's redirect-on-failure rule.

3.5 Pool v2: robustness levers shift the nominal optimum (amendment 2026-09-26)

Following the redirect rule, pool v2 was locked as a dated amendment before evaluation: four host-level robustness levers - acetate-overflow knockout (ackA-pta), formate-overflow knockout (pflB), lactate-overflow knockout (ldhA), and a glycerol co-utilization feed - added to the target-specific blocks, expanding the genome space to 64 genomes per target. Exhaustive enumeration (192 genomes total, metric-independent of the pending battery redesign) was again strictly dominant over the stochastic budget at this size. Under the nominal condition at $g = 0.5$, the co-utilization feed dominates every target: best nominal fluxes rise to 6.496 (1,4-BDO), 6.498 (isobutanol), and 0.730 (lycopene) mmol/gDW/h, against envelope values of 5.267, 5.239, and 0.488 - nominal-condition exceedances of 23%, 24%, and 50% over the published routes evaluated in the same model. The gain is attributable to the added carbon supply, so the honest comparison under co-feeding must re-baseline the benchmark route with the same feed block enabled:

$$E_a^{\text{cofeed}} = \frac{F_a^{(\text{cofeed})}(0.5)}{F_{\text{benchmark}}^{(\text{cofeed})}(0.5)}, \quad (10)$$

with both sides evaluated under identical uptake bounds including the glycerol co-feed. That re-baselined comparison has now been executed (amendment 2026-09-27 10:04): $E^{\text{cofeed}} = 1.00$ on all three targets (6.496 vs 6.496; 6.498 vs 6.498; 0.730 vs 0.730) - the nominal exceedance is entirely attributable to the added carbon supply, and no nominal-flux advantage over the published routes survives equal-feed comparison. The 1000-draw random-architecture null likewise recovers the pool optimum on all three targets, confirming that nominal-condition search is trivial at this pool size; the project's discriminating content is the robustness structure below, not nominal flux. An honest secondary negative is recorded: for isobutanol, some knockout combinations abolish product flux entirely (minimum 0.0), so the levers are not uniformly benign even at nominal condition.

3.6 Tiered battery executed: satisfiability restored, H3 advantage unearned (amendment 2026-09-27)

The battery was redesigned and locked as a dated amendment on 2026-09-27 before evaluation: mild perturbations (glucose $\pm 20\%$, oxygen -50%) are scored as absolute retention versus nominal with a hard leg of ≥ 0.8 each, and severe nutrient shifts (GLC_LOW, GLC_MID, ANAEROBIC, GLYCEROL, ACETATE) are scored against the pool frontier in the same condition,

$r_k = v_{p,k} / \max_a v_{p,k}(a)$, with a hard leg of ≥ 0.8 each; $R_{\text{tiered}} = 0.5 R_{\text{pert}} + 0.5 R_{\text{shift}}$. The bar is now satisfiable: 16/64 (1,4-BDO), 18/64 (isobutanol), and 18/64 (lycopene) genomes pass both hard legs. The best designs score $R_{\text{tiered}} = 1.0$ in all three targets: pflB knockout plus glycerol co-feed plus the succinyl-CoA route at 6.496 mmol/gDW/h nominal (1,4-BDO), pflB knockout plus co-feed plus the NADH alcohol dehydrogenase at 6.497 (isobutanol), and co-feed plus idi push at 0.730 (lycopene). Two findings follow. First, the tail-only H2 candidate does *not* survive the tiered battery ($R_{\text{tiered}} = 0.8023$ under the corrected-semantics re-evaluation below, superseding the GLPK-era 0.768 by explicit diff; severe-condition frontier-relative retention 0.41–0.67 on GLC_LOW, GLC_MID, ANAEROBIC, and ACETATE, confirmed unchanged): its minimal-architecture claim stands at the nominal condition only, and the battery-robust designs are co-fed, not enzyme-minimal. Second, the H3 comparison at this pool size is corrected from the GLPK-era record: the superseded battery recorded the best-by-nominal and best-by- R_{tiered} genomes as coinciding in all three targets (3/3), reported at the time as an honest negative. The deterministic re-evaluation below finds coincidence in one of three targets (lycopene only): at 1,4-BDO and isobutanol the nominal-best genome ties the battery-best at the nominal condition (6.4956 and 6.4975 mmol/gDW/h, identical at full printed precision) but is not battery-optimal ($R_{\text{tiered}} = 0.9492$ and 0.9923 versus 1.0). At pool 64 a nominal-only search therefore misses the robust optimum in two of three targets at tied nominal fitness - tie-level evidence in the direction of the robust-selection principle, explicitly superseding the earlier honest-negative framing; the effect is a tie-break, not a flux gap. The queued resolution was executed under the 2026-09-27 10:04 amendment: pool v3 (two further host levers, adhE and tpiA knockouts; 256 genomes per target, exhaustive) again yields $\text{argmax}(v_{\text{nominal}}) = \text{argmax}(R_{\text{tiered}})$ in all three targets (rank-1 difference 0.0000 everywhere), so the H3 search-mode advantage is not earned at N=256 either - a second falsification, reported as such. Battery pass rates at N=256: 32/256 (1,4-BDO), 60/256 (isobutanol), 22/256 (lycopene). The same amendment's H2 redirect (Robust Minimal Architecture: minimum heterologous enzyme count K_{het} subject to battery pass, published anchor $K_{\text{het, Yim}} = 5$) is also falsified on pool v3: no battery-passing 1,4-BDO design with $K_{\text{het}} \leq 4$ exists - every robust design requires an upstream route block, so the published route's enzyme count is not beatable under the robustness constraint in this pool. The remaining live test of the robustness principle is the held-out-target arm (3-HP), locked in the same amendment.

3.7 Objective ablation (amendment 2026-09-27 12:37 IST, locked before computation)

Scoring-component ablation on the committed exhaustive pool-v3 results (the EA itself is inapplicable to ablate at this pool size - the space is enumerated 256/256 - so the operative units are the objective's components). Four re-rankings were compared per target: full R_{tiered} , nominal-flux only (robustness removed), R_{pert} only, and R_{shift} only. On all three targets all four arms select the *same* winning design; the severe-retention loss of the nominal-only ranking is exactly 0.0. Recorded verbatim per the locked pass rule: the robustness objective is non-binding at pool v3 - robustness and nominal flux are co-maximized by the same architectures on this pool. This is disclosed as a pool property; on larger pools where the trade-off binds, the objective can matter, and the claim made here is only the measured one. Diversity and constraint-penalty ablations are deferred with disclosure (EA-search components, not defined for exhaustive evaluation).

3.8 Unrestricted versus biologically constrained search (amendment 2026-09-27 12:38 IST, locked before computation)

Dropping the growth constraint and maximizing product flux directly over the same pool yields, per target, a measured statement of what the constraint buys. On 14BDO the unrestricted winner reaches 7.75 mmol/gDW/h against the constrained best's 6.50 (ratio 0.838) and is growth-viable, but it fails the committed battery outright (worst severe-condition retention 0.46): the unconstrained optimum overpromises at nominal and collapses under condition shifts, which is the realism function of the constrained objective in one number. On isobutanol the unrestricted winner is itself battery-robust (0.91), so the constraint is non-binding there; on lycopene the two optima agree within solver tolerance (ratio 1.011, recorded rather than smoothed). No unrestricted top-ten design is zero-growth on any target. The constraints therefore cost nominal flux only where they also buy measured robustness - a target-dependent answer, reported verbatim.

3.9 What the search is, precisely (queue item 16; definitional pass)

To remove any vagueness about the computational method: the *representation* is a binary genome over six host-lever blocks and two target-route blocks (256 architectures per target); the *search policy* at pool v3 is exhaustive enumeration of that space - the earlier $(\mu + \lambda)$ evolution strategy was retired once enumeration became strictly dominant, so no stochastic search enters any reported number; the *objective* is the locked tiered robustness score $R_{\text{tiered}} = 0.5R_{\text{pert}} + 0.5R_{\text{shift}}$ evaluated by constraint-based flux analysis at 50% condition-specific growth; and the *learning component* is nil - nothing in this pipeline learns from data. It is exact enumeration with a pre-registered scoring rule, and the paper's claims are limited to what that procedure supports.

3.10 Prioritized experimental shortlist (queue item 17; candidates only, no experiments)

The results yield a ranked, falsifiable validation program. (1) *14BDO constrained-versus-unrestricted A/B*: implement both the R_{tiered} winner (ko_pflB + cofeed_glycerol + route B) and the unrestricted winner (cofeed only, no route block); the framework predicts the unrestricted design retains < 0.5 of nominal production under GLC_LOW/ANAEROBIC while the constrained design retains ≥ 0.8 - a two-strain experiment that can kill the approach. (2) *Isobutanol winner confirmation*: the same architecture class is battery-robust in silico (worst severe 0.91); a single-condition-shifts panel on one constructed strain tests the tiered battery's severe-leg calibration directly. (3) *Lycopene non-binding check*: constrained and unrestricted optima coincide within tolerance, so lycopene is the negative-control arm - the prediction is no measurable robustness difference. Ordering is by expected information per construct: (1) tests the framework's central claim, (2) calibrates its battery, (3) controls for over-crediting the constraint.

3.11 Pool v4: a five-arm evolutionary operator race at $N = 16,384$ (amendments 4a71551, eede3fd, 2026-09-28)

The pool-v4 expansion adds six further host levers (ppc, gnd, pgi, me1, acs, icdhyr knockouts) to pool v3's six, giving twelve host levers plus two target-specific route levers: $2^{14} = 16,384$

genomes per target. (An early amendment draft mislabeled this as $2^{16} = 65,536$ by double-counting pool v3's two target levers; the error was caught and corrected in PRE-REGISTRATION.md before any result depended on it, and the executed design is the intended one.) At this size enumeration no longer dominates the stochastic budget by the locked margin, so the pool is the first on which the evolutionary algorithm itself is the instrument - and the first on which search *operators* can be compared meaningfully. The locked experiment is an operator race: five arms share the $(\mu + \lambda)$ configuration ($\mu = 24$, $\lambda = 48$, 60 generations, tournament $k = 3$, bit-flip $p = 1/n$, 2,904 evaluations per run, 17.7% of the space) and differ in one operator each - A0 baseline; A1 constraint-aware mutation (re-sample until unseen); A2 diversity tie-break in tournament selection; A3 adaptive mutation rate on stagnation; A4 surrogate-assisted selection (random forest, 50 trees, default hyperparameters, trained on the run's own evaluation history). The primary cohort is 75 runs (5 locked seeds $\times$ 5 arms $\times$ 3 targets) with a Mann-Whitney gate (arm versus A0 on best flux, exact two-sided, $p < 0.05$) locked outcome-blind, a 10,000-label permutation sensitivity pooling a labeled 8-run extension cohort that is never mixed into the primary gate, and the final tiered battery on every run's best genome with the severe frontier computed within the evaluated set (disclosed: 25 primary best-genomes per target, not the full pool). Two engineering records belong to the methods rather than the results: the implementation checkpoints every generation so that a killed run resumes bit-exactly (verified by interrupted-versus-uninterrupted equivalence), and a GLPK presolve variant was tested as a solver-speed candidate and rejected after 1646/2568 re-evaluated fitnesses mismatched the committed values (maximum relative difference 13.45) - the solver configuration is part of the locked protocol, not a tuning knob.

The nominal optimum is flat. All 75 primary runs - every arm, every seed - converge to the same best flux per target: 13.4545 (1,4-BDO), 12.3333 (isobutanol), and 1.3704 (lycopene) mmol/gDW/h, roughly double the pool-v3 optima, so the six new host levers raise the nominal ceiling substantially, but no arm finds more of it than any other. The locked gate fires three times: isobutanol A1 ($p = 0.0079$), lycopene A2 ($p = 0.0079$), lycopene A3 ($p = 0.0079$), with the permutation sensitivity in agreement ($p = 0.0002, 0.0078, 0.0077$); all remaining cells sit at $p = 0.31$ – 1.0 . These passes are reported verbatim per the lock - and immediately qualified: the differences they detect are at the tenth to thirteenth significant digit of the flux (relative effect size $\sim 10^{-11}$), which is GLPK tie-breaking noise among equally optimal genomes, not an operator effect. Statistically decidable, scientifically zero. No arm produces a materially higher nominal flux than the baseline on any target, and at a flat optimum nominal fitness cannot in principle rank operators.

The battery separates what nominal fitness cannot - SUPERSEDED 2026-09-28 (see next subsection). The battery numbers in this paragraph were computed with the GLPK-based metric later shown to carry solver-state artifacts; they are retained verbatim for the audit trail and are replaced, number for number, by the repaired HiGHS metric below. The original text follows unchanged: Zero of the 75 best-by-nominal designs pass the full tiered battery (both hard legs at 0.8 in every condition): nominal-optimal is not robust, measured at pool-v4 scale. The arms nonetheless differ in *which* plateau genome they return, and therefore in its robustness (mean/max R_{tiered} over the five seeds): on 1,4-BDO, A0 scores 0.167/0.833 - its one design is the only genome among all 75 that passes the severe leg - while A1–A4 all score 0.000/0.000; on isobutanol, A0 0.235/0.393, A1 0.257/0.779, A2 0.375/0.612, A3 0.276/0.686, A4 0.161/0.324; on lycopene, A0 0.020/0.100, A1 0.100/0.200, A2 0.040/0.200, A3 0.120/0.200, A4 0.236/0.540. Tie-breaking among equally optimal genomes is not neutral: it moves robustness from 0.000 to 0.833 on the same fitness plateau, in both directions (the baseline lucked into the robust 1,4-BDO design; the surrogate arm picked the most robust

lycopene design). The measured lesson of the operator race is therefore twofold: nominal-condition search at this pool is a solved, flat problem for every operator tested, and the discriminating regime for operator design is robustness-aware selection - the plateau is wide, the plateau members differ enormously in battery robustness, and which member an operator returns is a designable property. A follow-up amendment searching directly under robust fitness (or a robustness tie-break at nominal ties) is the natural next experiment; it is proposed, not yet pre-registered, and no outcome under it has been touched.

3.12 Solver artifact in the tiered battery: detection, repair, and explicit supersession (amendments 2026-09-28 16:18–16:46 IST, each locked before its computation)

While instrumenting the A5 follow-up experiment below, the pool-v4 battery evaluation was found to depend on GLPK warm-start state rather than only on the LP being solved. The proof is direct: the search fitness and the battery’s nominal GLC_AEROBIC leg solve the *identical* LP (same base bounds, same $0.5\mu_{\max}$ floor), yet the committed battery recorded nominal 0.0 for four 1,4-BDO best genomes whose recorded search fitness is 13.4545; live, the same genome evaluated 13.4545 on a cold solver, 0.0 after a solver reset, and drifted after roughly one hundred warm evaluations. An interim repair (maximum of three perturbed GLPK solves per LP, locked 16:22 IST) reduced but did not eliminate the failures - 762 of roughly 2,000 genomes still failed self-validation - and was itself superseded with disclosure. The final repaired metric (locked 16:46 IST) exports each LP to matrices and solves it with HiGHS via `scipy.optimize.linprog`, which carries no state between solves; the metric is deterministic. The validation rule was locked before the recompute: for every evaluated genome the battery nominal must equal the recorded search fitness within 10^{-9} relative. **Result: zero failures across approximately 1,706 genomes** (83 committed best genomes plus 1,623 A5 plateau genomes).

Every battery-derived number in the preceding subsection is superseded; Table 2 gives the diff explicitly, number for number. The corrected picture inverts the earlier reading. Under the artifact metric, zero of 83 committed best genomes passed the battery and the arms appeared to differ sharply in robustness; under the repaired metric, **70 of 83 pass** (1,4-BDO 26/26 evaluated rows at primary+extension, isobutanol 25/32, lycopene 19/25; primary-cohort per-arm pass counts in the table), and best-by-nominal designs are usually battery-robust. What the superseded account attributed to robustness differences between plateau genomes was largely solver variance. The severe frontier itself moved only on ACETATE (1,4-BDO 6.1818 $\rightarrow$ 10.0; isobutanol 5.6667 $\rightarrow$ 9.1667; lycopene 0.6296 $\rightarrow$ 1.0185) - the artifact was concentrated in the per-genome evaluations, not the frontier. The “lucky A0 genome” narrative is corrected as well: A0’s 1,4-BDO designs were not barely-lucky survivors, they already sit at the robust optimum ($R_{\text{tiered}} = 1.277$, severe-leg parity with the evaluated-set frontier on all five legs and mild-leg retentions 1.39–1.72), and so do nearly all arms’ best genomes on all three targets.

3.13 Pool-v2/v3 re-evaluation under corrected semantics, and solver documentation (amendments dbd0b8c and 70a6872, 2026-09-28 18:14 and 18:50 IST, each locked before its computation)

The pool-v2/v3 batteries were computed under the same GLPK carried-state pipeline shown artifact-prone at pool v4, so a deterministic re-evaluation was locked as record integrity. The

Table 2: Explicit supersession diff, pool-v4 battery (primary cohort, 5 seeds per arm): mean/max R_{tiered} and full-battery passes under the superseded GLPK metric versus the repaired deterministic HiGHS metric (results/battery_v4_* vs results/battery_v6.json).

Target	Arm	GLPK mean/max	GLPK pass	HiGHS mean/max	HiGHS pass
1,4-BDO	A0	0.167/0.833	0/5	1.277/1.277	5/5
1,4-BDO	A1–A4 (each)	0.000/0.000	0/5	1.277/1.277	5/5
isobutanol	A0	0.235/0.393	0/5	1.228/1.272	4/5
isobutanol	A1	0.257/0.779	0/5	1.210/1.270	4/5
isobutanol	A2	0.375/0.612	0/5	1.227/1.277	4/5
isobutanol	A3	0.276/0.686	0/5	1.265/1.277	5/5
isobutanol	A4	0.161/0.324	0/5	1.221/1.272	4/5
lycopene	A0	0.020/0.100	0/5	1.174/1.277	2/5
lycopene	A1	0.100/0.200	0/5	1.208/1.277	3/5
lycopene	A2	0.040/0.200	0/5	1.243/1.277	4/5
lycopene	A3	0.120/0.200	0/5	1.277/1.277	5/5
lycopene	A4	0.236/0.540	0/5	1.277/1.277	5/5

first execution (dbd0b8c) contained an adaptation defect, disclosed verbatim: the script omitted the original metric’s biomass unpin, so its growth LP only tested the feasibility of a pre-pinned $0.5 \mu_{\text{max}}$ bound - three severe legs (GLC_LOW, ANAEROBIC, ACETATE) returned artifactual zeros for every pool-target, and the remaining legs used a $0.25 \mu_{\text{max}}$ floor rather than 0.5 times the condition-specific maximum. A hand check ordered before any paper fold-in reproduced the artifact exactly; the execution was withdrawn in full, its outputs retained as audit trail, and the interim reading that the committed pool numbers were artifact-inflated was retracted. Nothing from the defective execution entered this paper. The corrected execution (70a6872) replicates the original semantics exactly - unpin biomass, true condition-specific maximum growth, $0.5 g$ floor, then product - under deterministic HiGHS, validated before use by (a) independent fresh-process re-derivation of 54 leg values (6 pool-targets $\times$ 9 conditions, independent code, all within 10^{-9} relative), (b) all 15 comparable pool-v2 nominals matching the locked reference exactly (the reference file proved to hold only 6 rows per target; the coverage gap is disclosed), (c) all 768 pool-v3 nominals matching the committed GLPK-era values within 10^{-9} relative, and (d) deterministic repeat solves. Outcome: every published pool-v2/v3 number re-stands - pass counts (16/64, 18/64, 18/64; 32/256, 60/256, 22/256), severe frontiers (no leg differing by more than 5×10^{-3}), best R_{tiered} , and pool-v3 pass sets (symmetric difference zero) - with exactly three exceptions, each corrected explicitly here: the two tie-level argmax-coincidence flags at pool 64 and the tail-only candidate’s R_{tiered} ($0.768 \rightarrow 0.8023$). The defect, its detection, and its withdrawal remain in the record deliberately: a traceable correction history is part of the result.

Solver documentation (external-review requirements, adopted verbatim). Search fitness and all superseded batteries used GLPK 5.0 via optlang 1.9.1; all current battery numbers use HiGHS via `scipy.optimize.linprog` (scipy 1.15.3, method=’highs’, default tolerances) on exported LPs: the stoichiometric matrix from cobra 0.32.1, zero equality right-hand side, and bounds read from the reaction objects after base-medium, condition, and genome application. A solver reset means a fresh independent `linprog` call per LP; no basis, objective, or model state is carried between any two solves, and genome application is rolled back per genome by context manager. The locked validation algorithm is: for every evaluated genome, solve the nominal-condition product LP under the $0.5 g$ floor and require agreement with that

genome’s recorded search fitness within 10^{-9} relative before any battery value is used; every mismatch is disclosed verbatim. Fresh-process reproduction with independent code was performed for both the cohort battery and the pool re-evaluation (54/54 legs each, tolerance 10^{-9}). Superseded values never enter current statistics, figures, or denominators; they appear only as explicitly labeled historical audit data.

3.14 A5: trajectory-logged plateau selection converts plateau luck into a guarantee (amendments 5c99318, 4dfa860, 25b9007; locked before computation)

The operator race showed a flat nominal optimum whose plateau members were expected to differ in robustness; the repaired metric confirms the premise and corrects the scale. The A5 arm was locked as a two-stage follow-up on the same 15-run grid (5 seeds reused for clean comparability, single arm; a direct R_{tiered} -as-fitness variant was rejected on $\sim 9\times$ evaluation cost). Stage 1 replays the A0 search with full trajectory logging - every explored genome and fitness, satisfying the instrumentation mandate - and is verified bit-equivalent to the committed A0 runs (per-run best flux matched within 10^{-9} relative; the one strict-equality miss, isobutanol seed 261927 at 5×10^{-13} relative, is GLPK tie-break noise between cold and warm model states, disclosed in the amendment trail). Stage 2 applies a locked selection rule: the *plateau* (all explored genomes within 10^{-6} relative of the run’s best flux, 77–154 members per run) is battery-evaluated under the repaired metric, and the design with maximal R_{tiered} is returned (ties broken by first exploration).

A5 selects a frontier-parity robust design in 15 of 15 runs. Every A5 selection passes the full tiered battery at $R_{\text{tiered}} = 1.277$ - severe-leg parity with the evaluated-set frontier on all five legs, mild-leg retentions 1.39–1.72. The locked gate (Mann–Whitney exact, A5-selected versus the committed A0 cohort’s repaired R_{tiered} , 5v5 per target, with 10,000-label permutation sensitivity) fires at $p = 0.0079$ on all three targets - and the effect-size audit, applied exactly as it was to the nominal-flux gate, splits them: on 1,4-BDO the A5–A0 differences are $\sim 10^{-13}$ (floating-point tie-break epsilon among equally robust-optimal genomes; the gate fires on ranks of identical numbers, statistically decidable, scientifically null, disclosed verbatim); on isobutanol the differences are 0.005–0.171, all positive, permutation $p = 0.0023$ - **a substantive computational effect** (a model-property result, not a biological effect); on lycopene the differences are 0.171 on three seeds and epsilon-tied on two, permutation $p = 0.0033$ - heterogeneous benefit, genuine on three of five seeds, not a uniform improvement. The substantive claim is reliability, not a higher ceiling: the robust optimum coincides with the nominal optimum on this pool (Section above), so an operator that merely tie-breaks on nominal fitness reaches it only by luck - the committed A0 cohort attains the exact robust optimum on 0/5 (isobutanol), 2/5 (lycopene), and 5/5-by-epsilon (1,4-BDO) seeds - while trajectory logging plus plateau selection attains it on 15/15. Robustness-aware final selection is the measured fix for the tie-break neutrality question the operator race posed.

Tie-break control: where does selection beat luck? (amendment 10faf32, locked before inspection). External review noted that 70 of 83 repaired best genomes pass the battery, so A5’s 15/15 could reflect a favorable pool rather than a selection-rule contribution. The locked control compares, per run, the A5-selected R_{tiered} with the run’s full plateau distribution: f_{tie} is the probability that a uniform-random plateau tie-break attains the A5-selected value. By the locked rule (win at median $f_{\text{tie}} \leq 0.2$) the control returns a **null overall**: median $f_{\text{tie}} = 0.817$

across the 15 runs - in most runs any tie-break finds a frontier-parity design. The per-target decomposition mirrors the effect-size audit exactly. At 1,4-BDO $f_{\text{tie}} = 1.0$ in all five runs: every plateau member is frontier-parity, selection contributes nothing, and the realized A0 luck selections sit at the same value. At lycopene the median $f_{\text{tie}} = 0.82$ - mostly luck. At isobutanol the median $f_{\text{tie}} = 0.098$ (range 0.017-0.252, A5 percentiles 87-99, realized A0 selections at plateau percentiles 18-74): robustness-aware selection beats a random tie-break decisively. The three targets are therefore not three equivalent successes: 1,4-BDO is an epsilon-level null, isobutanol a substantive computational effect, and lycopene a heterogeneous benefit on three of five seeds - and the tie-break control shows the same asymmetry in the selection mechanism itself. A5's measured contribution is real but target-dependent, not a uniform guarantee.

4 Caveats

(1) All results are in-silico flux predictions; no wet-lab validation is claimed, and no build-phase engineering guidance is produced. (2) Both pools evaluated so far are deliberately small; the exhaustive results bound what these pools can express, and conclusions about search power await larger pools. In particular, the H3 multi-condition advantage was first recorded as undetectable at pool 64 under the superseded GLPK-era battery; the deterministic re-evaluation finds the nominal-best and battery-best designs coincide in only one of three targets there, at tied nominal fitness, so pool-64 evidence is tie-level and pool-256 remains the falsifying scale. (3) The tail-only H2 candidate is verified at the nominal condition only: it fails the tiered battery, every battery-passing design requires the glycerol co-feed, and under equal-feed comparison ($E^{\text{cofeed}} = 1.00$) the co-feed fully accounts for the nominal-flux gains - no flux advantage over the published routes remains. The defensible claims are the robustness structure and its cross-target motif, not nominal flux. (4b) H3's multi-condition search advantage is falsified at N=256 (nominal-best and battery-best designs coincide in every target, rank-1 difference 0.0000); at N=64 the corrected-semantics re-evaluation reverses the superseded GLPK-era coincidence reading (coincidence in one of three targets, at tied nominal), so the earlier N=64 honest-negative framing is withdrawn. (5) At pool v4 the nominal optimum is flat and fully reached by every operator arm tested; operator claims are therefore limited to tie-breaking effects on battery robustness, and the three locked-gate passes are numerically significant solver-noise effects ($\sim 10^{-11}$ relative), reported verbatim and not counted as arm advantages. (6) At pool v4 the growth-coupling floor is non-binding: maximum growth under the locked base condition evaluates to numerical zero ($\sim 5 \times 10^{-31}$) for measured genomes, so the $0.5 \mu_{\text{max}}$ floor never constrains and all pool-v4 fluxes are effectively uncoupled maxima; mild-leg retentions above 1 (perturbed conditions yielding more product than nominal) are deterministic properties of this model's bound set, reported verbatim and not claimed as biology. (7) Battery solver history: the pool-v4 battery was first computed under GLPK with carried warm-start state (artifact, superseded), then under a max-of-3 perturbed GLPK repair (insufficient - 762 self-validation failures, superseded), and finally under the deterministic HiGHS metric with 100% nominal-versus-search validation across $\sim 1,706$ genomes; only the HiGHS numbers are current, and the superseded values remain in the record for audit. (8) The pool-v2/v3 batteries were computed under the same GLPK pipeline; they have since been re-evaluated under deterministic HiGHS with the original metric semantics (amendment 70a6872, locked before computation), after a first re-evaluation execution contained an adaptation defect - a missing biomass unpin that artifactually zeroed three severe legs - caught by a pre-fold-in hand check, withdrawn, and retained in the record. Under the corrected re-evaluation every published pool-v2/v3 number re-stands

except the two tie-level pool-64 coincidence flags and the tail-only R_{tiered} ($0.768 \rightarrow 0.8023$); all diffs appear explicitly in the text. (4) GSMM predictions assume steady state and fixed biomass composition; real titers are typically lower.

5 Next steps

(1) DONE: co-feed re-baseline ($E^{\text{cofeed}} = 1.00$ on all targets) and the 1000-draw random null (recovers the optimum; nominal search trivial at this size). (2) DONE under the 10:04 amendment: pool-v3 expansion and H2-RMA redirect, both falsified as recorded above. (3) Held-out 3-HP target arm (3B, locked 10:04): construct the malonyl-CoA reductase route, run the identical tiered battery on the 64-genome host-lever pool, Fisher exact test on co-feed + pflB motif enrichment. (4) Yeast8 cross-host arm for the robustness principle. (5) DONE (A5, this fold-in): trajectory-logged plateau selection under the repaired deterministic battery - genuine gated effects on isobutanol and lycopene, epsilon-null on 1,4-BDO disclosed. (6) PROPOSED, owner decision pending before any lock: a sixth search arm (robust-fitness search or robustness tie-break at nominal ties), plus a HiGHS re-evaluation of the pool-v2/v3 batteries under the repaired metric.

5.1 Biological realism screen (amendment 2026-09-27 12:32 IST, locked before application)

The top evolved designs were screened against two locked realism filters. F1 (enzyme burden): heterologous reaction count may not exceed the locked literature benchmark pathway's count by more than one; all top-ten designs by R_{tiered} pass on all three targets (benchmarks: 5, 5, and 4 heterologous steps), as do both per-target winners - the evolved architectures are parsimonious relative to demonstrated routes. F3 (cofactor-condition consistency): no top-ten design combines an NADPH-dependent active route block with zero anaerobic retention. F2 (ATP-cost screen) is deferred with explicit disclosure: per-design FBA solution vectors are not stored in the committed results, and computing them is queued as separate compute. No design was re-ranked; the screen is report-only.

5.2 Per-architecture rationale: why the battery winners win

Each battery-winning genome is interpretable block by block, and the interpretation is the same logic the tiered battery was designed to select for. **1,4-BDO winner** (ko_pflB + glycerol co-feed + succinyl-CoA route, nominal 6.496 mmol/gDW/h, $R_{\text{tiered}} = 0.994$): the pflB knockout removes the formate overflow arm, keeping pyruvate carbon inside the central network where the succinyl-CoA route can draw on it; the co-feed adds a second, more reduced carbon source whose uptake is uncoupled from glucose repression in the model. The design trades nothing at nominal and retains frontier-relative flux on every severe shift except anaerobiosis (0.938), where formate removal is nearly free because pyruvate formate-lyase is itself oxygen-sensitive. **Isobutanol winner** (ko_pflB + ko_adhE + co-feed + NADH-alcohol dehydrogenase, 6.498, $R_{\text{tiered}} = 0.9997$): the adhE knockout blocks the ethanol drain on acetyl-CoA and NADH, and choosing the NADH-dependent alcohol dehydrogenase matches the cofactor pool the knock-offs enrich - the architecture is internally redox-consistent, which is exactly the property the mild oxygen-limitation leg (retention 1.0) measures. **Lycopene winner** (co-feed + idi push, 0.730, $R_{\text{tiered}} = 1.0$, the only perfect score): the idi push balances IPP:DMAPP isomer supply

- the known bottleneck of the heterologous crtEBI tail - while the co-feed raises precursor flux without touching dxs; the design reaches the battery frontier in all nine conditions, including anaerobiosis, where its retention is 1.0. Common logic across all three: carbon supply is the binding lever - *every* battery-passing genome in all three targets carries the glycerol co-feed (32/32, 60/60, and 22/22 of passers), while no genome passes without it, and conservation knockouts are optional score-improving refinements rather than requirements (co-feed-only genomes pass in all three targets; the rank-1 designs pair the co-feed with a knockout in two of three). The design principle the battery surfaces at $N=256$ is therefore sharper than a generic conjunction: unlock carbon supply first, then conserve it. This is a statement about which levers robust designs share, not about a multi-condition search advantage - H3 was falsified at $N=256$ (argmax coincidence, above) and nothing here resurrects it.

5.3 Held-out target: 3-hydroxypropionate (3B)

The remaining live test of the robustness principle was run under its 10:04 IST lock: 3-HP via the heterologous MCR1/MCR2 route in the same iML1515 base, 64 host-lever genomes, identical tiered battery with the severe frontier computed within the 3-HP pool. Six of 64 genomes pass both hard legs (nominal max 10.91 mmol/gDW/h); none carries the locked motif $M = \{\text{co-feed, ko_pflB}\}$ (Fisher one-tailed $p = 1.0$), so H0,3B is not rejected and the specific winner architecture is falsified as a portable motif, per the locked rule with no loosening. The motif's decomposition is the finding: all six passers carry the glycerol co-feed (the supply-lever necessity replicates on a held-out chemistry), while none carries the pflB knockout (PFL carries anaerobic pyruvate flux the NADPH-demanding MCR route cannot spare). The transferable principle is carbon-supply unlock; knockout pairings are target-specific.

5.4 FBA-assumption sensitivity (queue #4)

The three battery winners were re-evaluated on a locked 12-cell grid (biomass fraction $\{0.4, 0.5, 0.6\} \times$ glycerol uptake $\{-2, -4\} \times$ glucose base $\{-8, -10\}$), surviving 6/12 (1,4-BDO), 8/12 (isobutanol), and 10/12 (lycopene) cells. The mild leg never fails (minimum mild retention 1.0 in all 36 cells); all failures sit in the stacked worst case (forced growth 0.6 with throttled feeds) on the severe leg. Robustness therefore holds under single-parameter perturbation and degrades gracefully, with the lycopene design most assumption-stable.

5.5 Monte Carlo FBA uncertainty (queue #15)

Under the 13:35 IST lock, each winner was re-evaluated over 200 draws with every battery uptake bound independently jittered by $U(0.8, 1.2)$ (seed 260927). Battery-pass fractions: 1,4-BDO 92.5% (mean $R_{\text{tiered}} = 0.993$, 5–95% [0.949, 1.043]), isobutanol 98.0% (0.999, [0.955, 1.049]), lycopene 100% (0.999, [0.965, 1.037]). Winner robustness is not a knife-edge property of the locked bounds; the 7.5% 1,4-BDO failure tail is reported as the sensitivity mass below the pass bar.

5.6 Cross-organism transfer: native-yeast arm (queue #11)

The carbon-supply principle was tested in Yeast8 (native succinate export, 50% growth, 7 conditions mirroring the battery; amendment 14:15 IST, all reaction IDs verified pre-lock). The

glycerol co-feed improves succinate flux in 6 of 7 conditions with the largest gain under carbon limitation (GLC_LOW 1.49 $\rightarrow$ 3.05 mmol/gDW/h, +104%) - but collapses anaerobically (6.81 $\rightarrow$ 0.00), consistent with yeast's oxygen-dependent glycerol catabolism, so the locked all-conditions prediction is reported as not met. Together with the 3-HP hold-out, the transferable content is the abstract principle (unlock carbon supply, gains largest under limitation); lever choice is host-specific.

5.7 Fourth chemical class: native succinate (queue #10)

A native organic-acid target was added under the 14:17 IST lock (64 host-lever genomes, identical battery). Ten of 64 genomes pass both hard legs (nominal max 10.66 mmol/gDW/h), and all ten carry the glycerol co-feed (Fisher one-tailed $p = 0.0004$, pre-locked test): the carbon-supply-necessity pattern replicates on a fourth, chemically distinct target class, its first replication under a pre-registered significance gate.